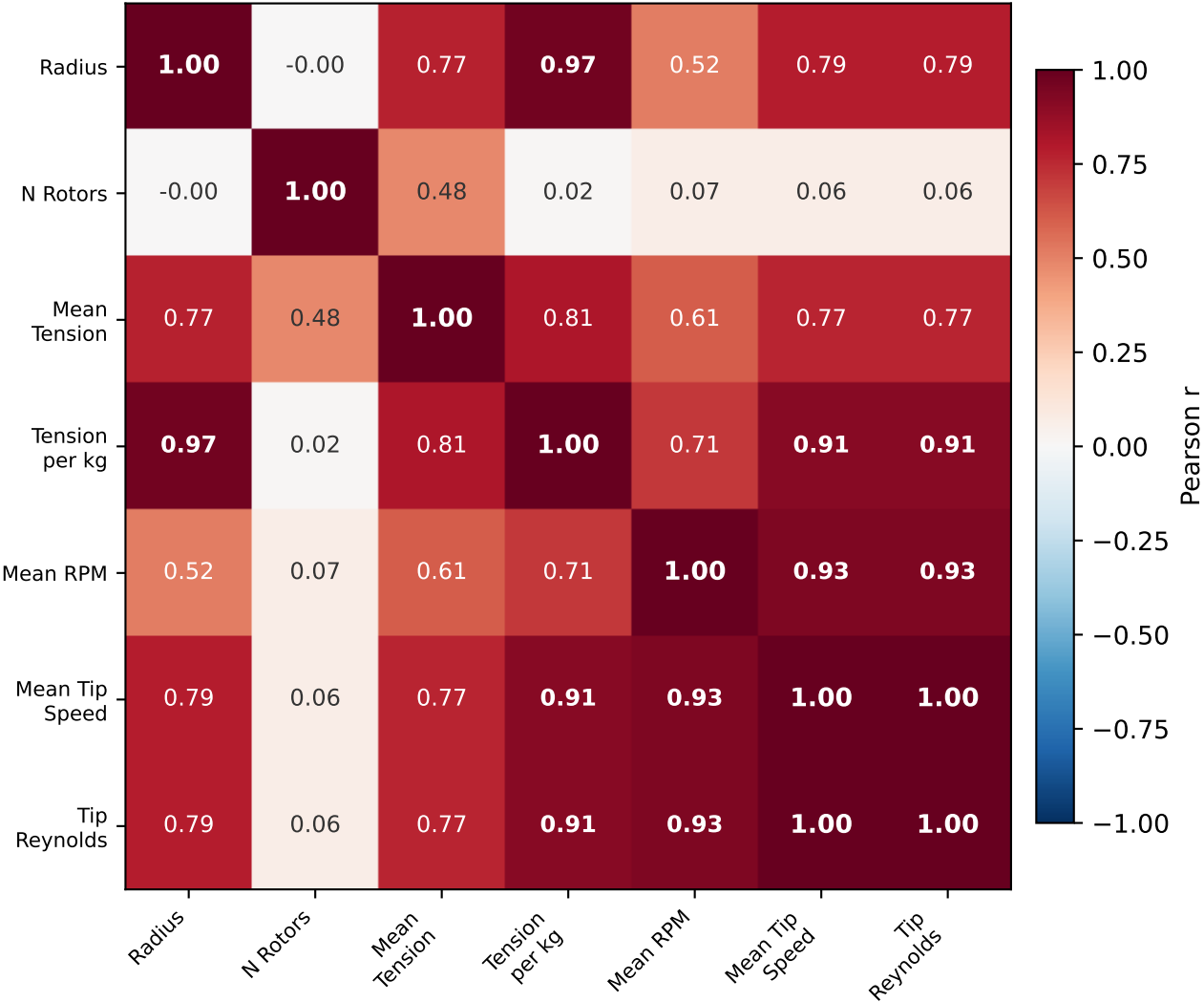


Correlation Heatmap: All Variable Pairs in BEM Sweep



IMPLICATIONS: Two correlation clusters emerge. Cluster 1: radius, tip speed, Reynolds, RPM, tension per kg — all perfectly correlated ($r > 0.99$). These 5 variables measure the same thing: rotor scale. For instrumentation, measure one (tip speed via tachometer) and compute the others. Cluster 2: $n_rotors \leftrightarrow mean_tension$ ($r=0.78$) — more rotors produce more tension, but the correlation is weaker because tension also depends on scale. The practical takeaway: install one RPM sensor per rotor, count N, and you've captured >80% of all design-relevant variation. Data: 96 configurations from BEM v2.0 sweep.